

How to make AI Superclusters

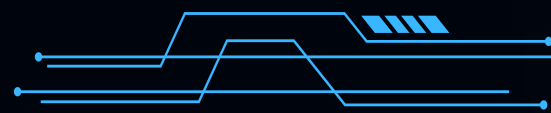




How to make AI Superclusters

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CTO of SIAM.AI CLOUD





We are a high-performance cloud computing provider specializing in AI Solutions.

“Driving Thailand to become an AI innovation hub in Southeast Asia”



How to make AI Superclusters

Contents

- High-performance AI Processing Technology
- High-performance Network Technology
- High-Performance Storage Technology
- Cooling system
- Network Design
- Data Centers

High-performance AI Processing Technology

- NVIDIA H100/H200 Hopper



How to make AI Superclusters

High-performance AI Processing Technology

- NVIDIA H100/H200 Hopper

Technical Specifications		
	H100 SXM	H100 NVL
FP64	34 teraFLOPS	30 teraFLOPS
FP64 Tensor Core	67 teraFLOPS	60 teraFLOPS
FP32	67 teraFLOPS	60 teraFLOPS
TF32 Tensor Core*	989 teraFLOPS	835 teraFLOPS
BFLOAT16 Tensor Core*	1,979 teraFLOPS	1,671 teraFLOPS
FP16 Tensor Core*	1,979 teraFLOPS	1,671 teraFLOPS
FP8 Tensor Core*	3,958 teraFLOPS	3,341 teraFLOPS
INT8 Tensor Core*	3,958 TOPS	3,341 TOPS

Technical Specifications		
	H100 SXM	H100 NVL
GPU Memory	80GB	94GB
GPU Memory Bandwidth	3.35TB/s	3.9TB/s
Decoders	7 NVDEC 7 JPEG	7 NVDEC 7 JPEG
Max Thermal Design Power (TDP)	Up to 700W (configurable)	350-400W (configurable)
Multi-Instance GPUs	Up to 7 MIGs @ 10GB each	Up to 7 MIGS @ 12GB each
Form Factor	SXM	PCIe dual-slot air-cooled
Interconnect	NVIDIA NVLink™: 900GB/s PCIe Gen5: 128GB/s	NVIDIA NVLink: 600GB/s PCIe Gen5: 128GB/s
Server Options	NVIDIA HGX H100 Partner and NVIDIA-Certified Systems™ with 4 or 8 GPUs NVIDIA DGX H100 with 8 GPUs	Partner and NVIDIA-Certified Systems with 1-8 GPUs

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High-performance AI Processing Technology

- NVIDIA DGX **Blackwell** B200



GPU	8x NVIDIA Blackwell GPUs
GPU Memory	1,440GB total GPU memory
Performance	72 petaFLOPS training and 144 petaFLOPS inference
Power Consumption	~14.3kW max
CPU	2 Intel® Xeon® Platinum 8570 Processors 112 Cores total, 2.1 GHz (Base), 4 GHz (Max Boost)
System Memory	Up to 4TB

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High-performance AI Processing Technology

- NVIDIA Grace Blackwell GB200 NVL36/NVL72

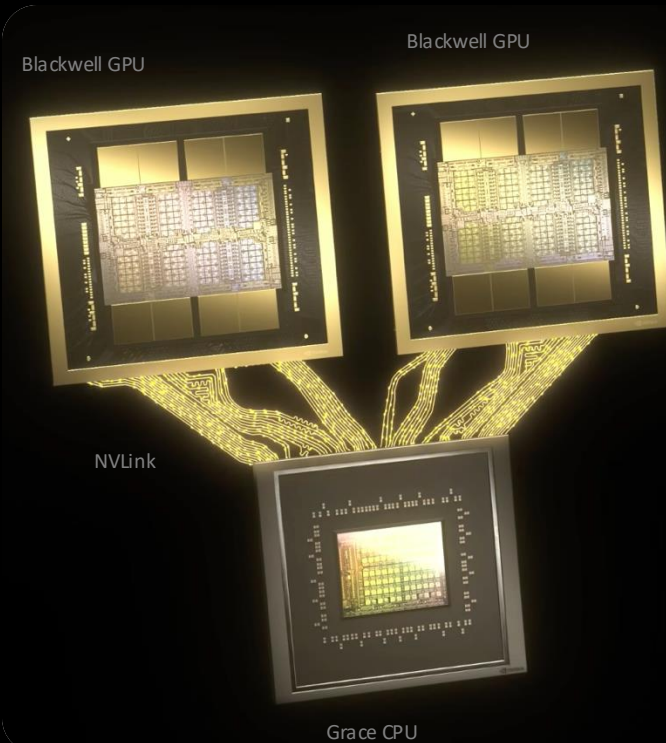


	GB200 NVL72	GB200 Grace Blackwell Superchip
Configuration	36 Grace CPU : 72 Blackwell GPUs	1 Grace CPU : 2 Blackwell GPU
FP4 Tensor Core ²	1,440 PFLOPS	40 PFLOPS
FP8/FP6 Tensor Core ²	720 PFLOPS	20 PFLOPS
INT8 Tensor Core ²	720 POPS	20 POPS
FP16/BF16 Tensor Core ²	360 PFLOPS	10 PFLOPS
TF32 Tensor Core	180 PFLOPS	5 PFLOPS
FP32	6,480 TFLOPS	180 TFLOPS
FP64	3,240 TFLOPS	90 TFLOPS
FP64 Tensor Core	3,240 TFLOPS	90 TFLOPS
GPU Memory Bandwidth	Up to 13.5 TB HBM3e 576 TB/s	Up to 384 GB HBM3e 16 TB/s
NVLink Bandwidth	130TB/s	3.6TB/s
CPU Core Count	2,592 Arm® Neoverse V2 cores	72 Arm Neoverse V2 cores

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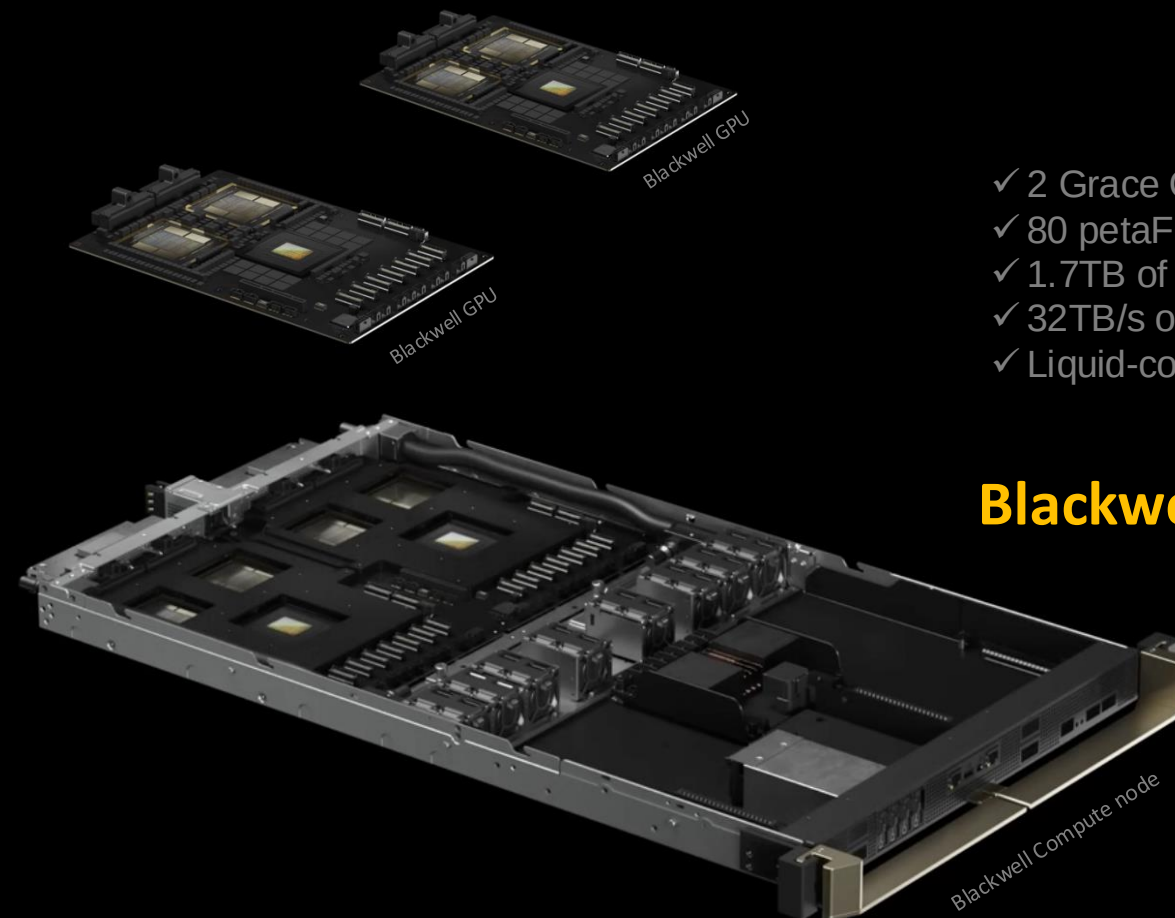
High-performance AI Processing Technology

- NVIDIA **Grace Blackwell** GB200 NVL36/NVL72



2 Blackwell GPUs and 1 Grace CPU

- ✓ 384GB of HBM3e
- ✓ 72 ARM NVIDIA Neoverse V2 CPU Cores
- ✓ 900GB/s of NVLink-C2C bandwidth



- ✓ 2 Grace CPUs and 4 Blackwell GPUs
- ✓ 80 petaFLOPS of AI performance
- ✓ 1.7TB of HBM3e
- ✓ 32TB/s of memory bandwidth
- ✓ Liquid-cooled MGX design

Blackwell Compute node

How to make AI Super clusters

High-performance AI Processing Technology

- NVIDIA **Grace Blackwell** GB200 NVL36/NVL72



GB200 NVL72

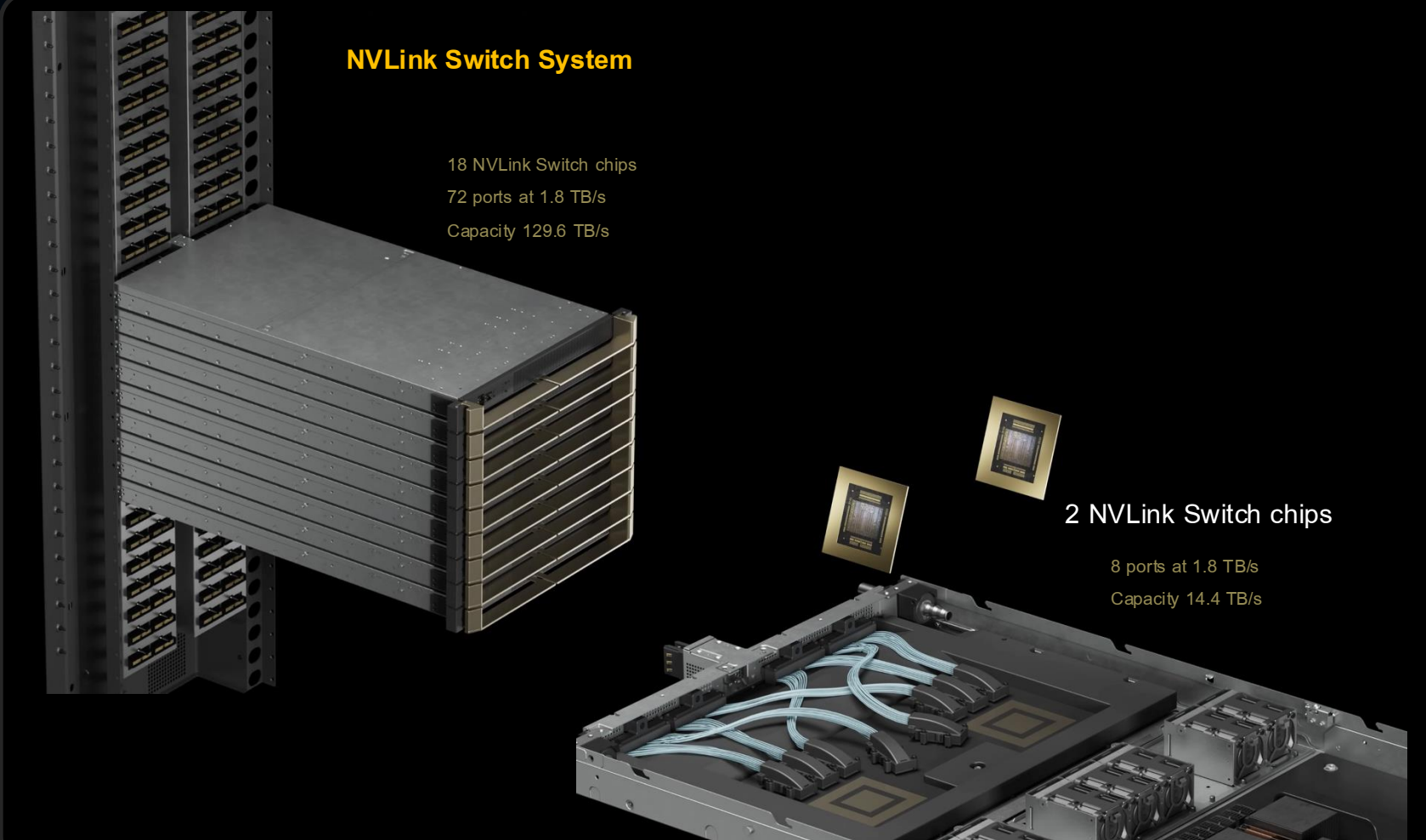
- ✓ 18 Compute Trays In A Rack
- ✓ 36 Grace CPUs And 72 Blackwell GPUs

COMPUTE FOR **TRILLION-PARAMETER-SCALE**
GENERATIVE AI

- ✓ 1.4 exaFLOPS of AI Performance
- ✓ 30TB of HBM3e

One billion equals 1,000,000,000

One trillion equals 1,000,000,000,000



NVLink Switch System

18 NVLink Switch chips
72 ports at 1.8 TB/s
Capacity 129.6 TB/s

2 NVLink Switch chips

8 ports at 1.8 TB/s
Capacity 14.4 TB/s

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High-performance Network Technology

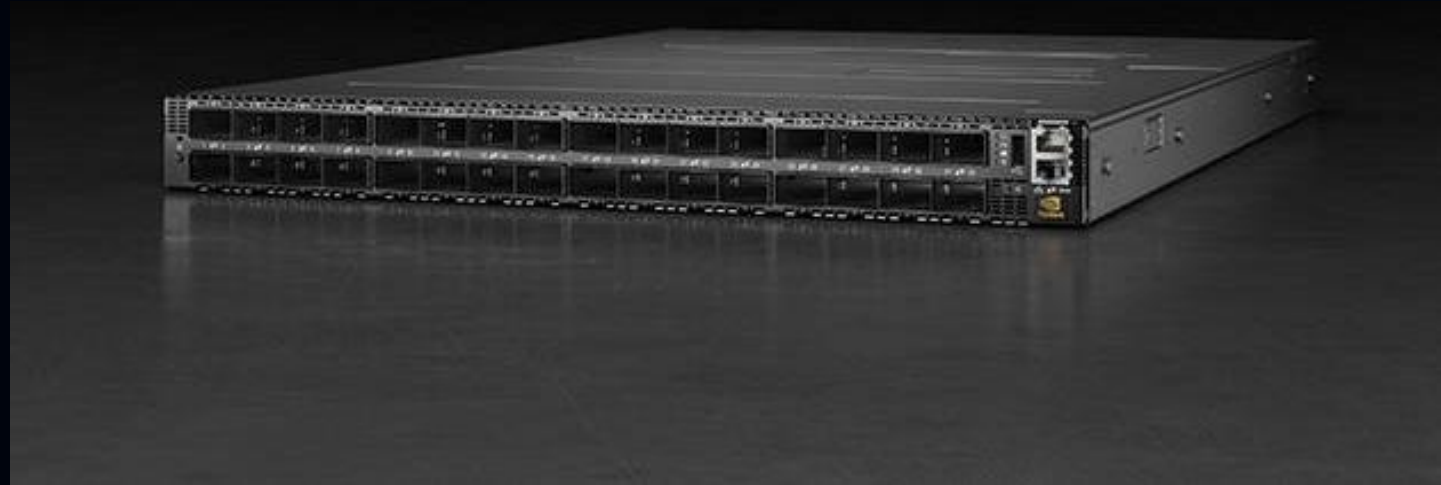


NVIDIA Spectrum Series Switches

- ✓ Up to **800GbE** per port for 51.2Tb/s aggregate switch bandwidth
- ✓ Accelerated **RDMA** over Converged Ethernet (RoCE) for AI, cloud, storage, and high-performance computing (HPC)
- ✓ Consistent and **low** cut-through **latency**

How to make AI Super clusters

High-performance Network Technology



NVIDIA Quantum-2 Series Switches

- ✓ NDR 400Gb/s InfiniBand per port
- ✓ NVIDIA InfiniBand
- ✓ Extremely low-latency

High-performance Network Technology



NVIDIA BlueField-3 Networking Platform

- ✓ 1 or 2 ports with up to **400Gb/s** connectivity
- ✓ 32GB on-board DDR5 memory
- ✓ 1GbE out-of-band management port
- ✓ Integrated BMC

How to make AI Super clusters

High-Performance Storage Technology

Delivering unprecedented throughput of up to **115 GB/s read and 75 GB/s write**, along with **3 million up to IOPS**



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High-Performance Storage Technology



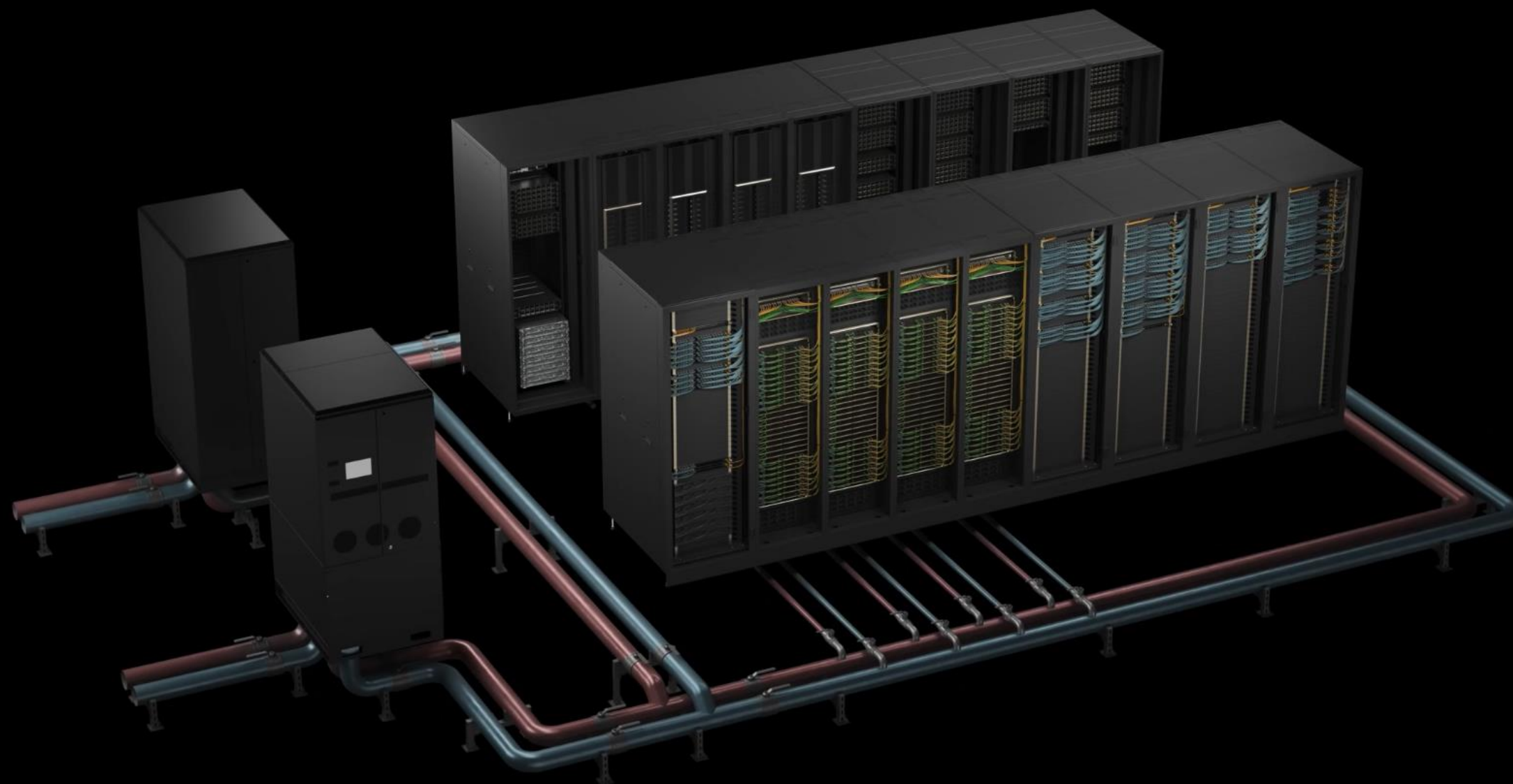
	Scalability	Initial Performance	Initial Capacity	Connectivity	Physical
Prime 120	Start with a minimum of 8 servers and scale to 100's of servers	120 GB/s Read BW 32 GB/s Write BW 3.6 Million IOPS	0.4PB Usable - PCIe Gen4 pTLC	16 x 200Gb Ethernet or InfiniBand (QSFP56)	8 Rack Units 3.6 kW 12 kBTU/hr
Prime 140	Start with a minimum of 8 servers and scale to 100's of servers	200 GB/s Read BW 56 GB/s Write BW 6.0 Million IOPS	0.7PB Usable - PCIe Gen4 pTLC	16 x 200Gb Ethernet or InfiniBand (QSFP56)	8 Rack Units 3.8 kW 13 kBTU/hr
Prime 160	Start with a minimum of 8 servers and scale to 100's of servers	320 GB/s Read BW 96 GB/s Write BW 12.0 Million IOPS	1.4PB Usable - PCIe Gen4 pTLC	16 x 200Gb Ethernet or InfiniBand (QSFP56)	8 Rack Units 4.3 kW 15 kBTU/hr
Nitro 150	Start with a minimum of 8 servers and scale to 100's of servers	720 GB/s Read BW 186 GB/s Write BW 18.3 Million IOPS	0.5PB Usable - PCIe Gen5 TLC	16 x 400Gb Ethernet or InfiniBand (OSFP)	8 Rack Units 6.4 kW 22 kBTU/hr
Nitro 170	Start with a minimum of 8 servers and scale to 100's of servers	720 GB/s Read BW 186 GB/s Write BW 18.3 Million IOPS	1.0PB Usable - PCIe Gen5 TLC	16 x 400Gb Ethernet or InfiniBand (OSFP)	8 Rack Units 6.4 kW 22 kBTU/hr

How to make AI Super clusters

Cooling system

GB200 NVL72s COMPUTE RACKS

Liquid Cooled For Best Energy Efficiency

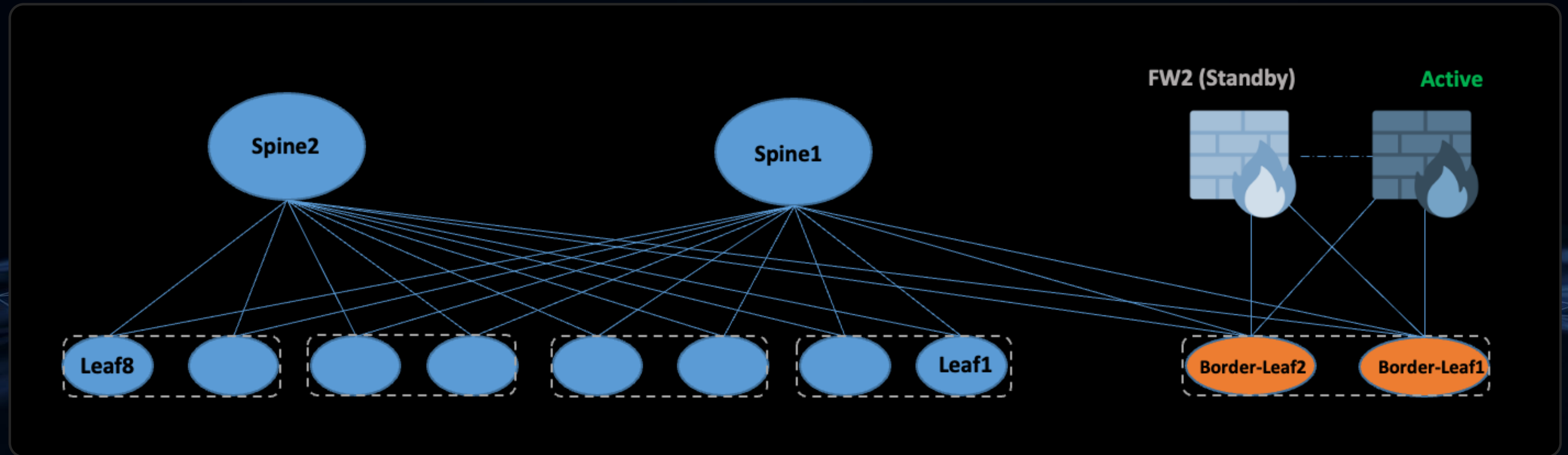


Network Design

Spine-Leaf Network Design

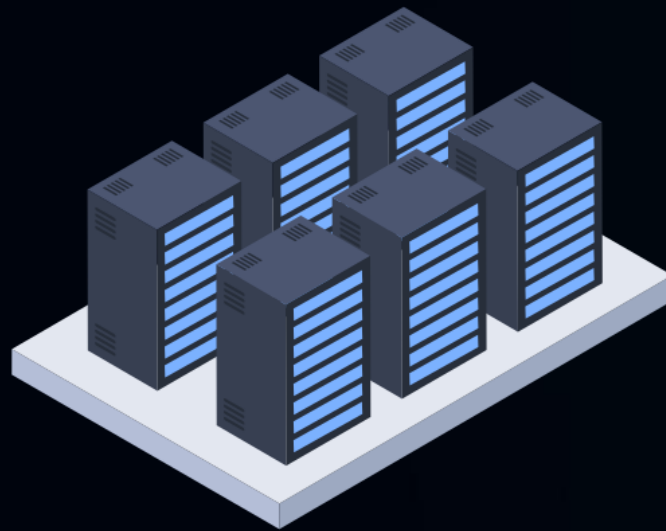
Dynamic Routing Protocol: OSPF, BGP

Redundant: MLAG, EVPN-Multi Homing



How to make AI Super clusters

Data Centers



Next Generation DC Requirement



Power Consumption



Air Flow, Air Cooling System

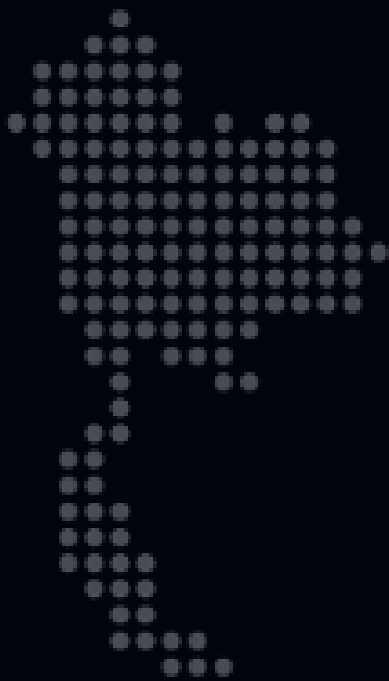


Liquid Cooling System

How to make AI Super cluster

THE LARGEST AI COMPUTING IN THAILAND

* FP64 Processing/GPU



Siam AI SuperNAP DC Cluster

34.81 PETAFL0P *

Cluster 1K (H100 x 1024)



Supercomputer (Private sector)

13.85 PETAFL0P



6.82 PETAFL0P *

The Government Cluster

How to make AI Super clusters

SIAM.AI GPU CLUSTER

Siam.AI has several high-performance computing clusters that can rank among the top in ASEAN and is currently expanding.



DC: SuperNAP

Hopper H100 (1K)

 34.81 PETAFL0P *
4,052.99 PETAFL0P **



DC: OneAsia

Hopper H200 (1K)

 34.81 PETAFL0P *
4,052.99 PETAFL0P **



DC: STT

Grace Blackwell GB200 (3.5K)

 155.52 PETAFL0P *
34,560 PETAFL0P **



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* FP64 Processing/GPU

** FP8 Processing/GPU



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